

Number systems and computers

- Computers store all data as binary digits, but we may need to convert this to a number system we are familiar with.
- Computer programs and data are often represented (outside the computer) using octal and hexadecimal number systems because they are a short hand way of representing binary numbers.

Data Representation

- Computers use digital representation
- Based on a binary system
(uses on/off states to represent 2 digits).
- Many different types of data.
- Examples?
- ALL data (no matter how complex)
must be represented in memory as binary digits (bits).

Introduction to Numbering Systems

- We are all familiar with the decimal number system (Base 10). Some other number systems that we will work with are
 - Binary $\rightarrow$ Base 2
 - Octal $\rightarrow$ Base 8
 - Hexadecimal $\rightarrow$ Base 16

Significant Digits

Binary: 1 1 1 0 1 1 0 1

Most significant bit

Least significant bit

Hexadecimal: 1D63A7A

Most significant digit

Least significant digit

Number Systems - Decimal

- The decimal system is a base-10 system.
- There are 10 distinct digits (0 to 9) to represent any quantity.
- For an n-digit number, the value that each digit represents depends on its weight or position.
- The weights are based on powers of 10.

$$1024 = 1 \cdot 10^3 + 0 \cdot 10^2 + 2 \cdot 10^1 + 4 \cdot 10^0 = 1000 + 20 + 4$$

Binary Number System

- Also called the “Base 2 system”
- The binary number system is used to model the series of electrical signals computers use to represent information
 - 0 represents the no voltage or an off state
 - 1 represents the presence of voltage or an on state

Decimal to Binary Conversion

- The easiest way to convert a decimal number to its binary equivalent is to use the **Division Algorithm**
- This method repeatedly divides a decimal number by 2 and records the quotient and remainder
 - The remainder digits (a sequence of zeros and ones) form the binary equivalent in least significant to most significant digit sequence

Division Algorithm

Convert 67 to its binary equivalent:

$$67_{10} = x_2$$

Step 1: $67 / 2 = 33 \text{ R } 1$

Divide 67 by 2. Record quotient in next row

Step 2: $33 / 2 = 16 \text{ R } 1$

Again divide by 2; record quotient in next row

Step 3: $16 / 2 = 8 \text{ R } 0$

Repeat again

Step 4: $8 / 2 = 4 \text{ R } 0$

Repeat again

Step 5: $4 / 2 = 2 \text{ R } 0$

Repeat again

Step 6: $2 / 2 = 1 \text{ R } 0$

Repeat again

Step 7: $1 / 2 = 0 \text{ R } 1$

STOP when quotient equals 0

1 0 0 0 0 1 1₂

Binary to Decimal Conversion

- The easiest method for converting a binary number to its decimal equivalent is to use the **Multiplication Algorithm**
- Multiply the binary digits by increasing powers of two, starting from the right
- Then, to find the decimal number equivalent, sum those products

Multiplication Algorithm

Convert $(10101101)_2$ to its decimal equivalent:

Binary

→ 1 0 1 0 1 1 0 1

X X X X X X X X

Positional Values

→ 2^7 2^6 2^5 2^4 2^3 2^2 2^1 2^0

Products

→ 128 + 32 + 8 + 4 + 1

173₁₀

Octal Number System

- Also known as the **Base 8 System**
- Uses digits **0 – 7**
- Readily converts to binary
- Groups of three (binary) digits can be used to represent each octal digit
- Also uses multiplication and division algorithms for conversion to and from base 10

Decimal to Octal Conversion

Convert 427_{10} to its octal equivalent.

$$427 / 8 = 53 \text{ R}3$$

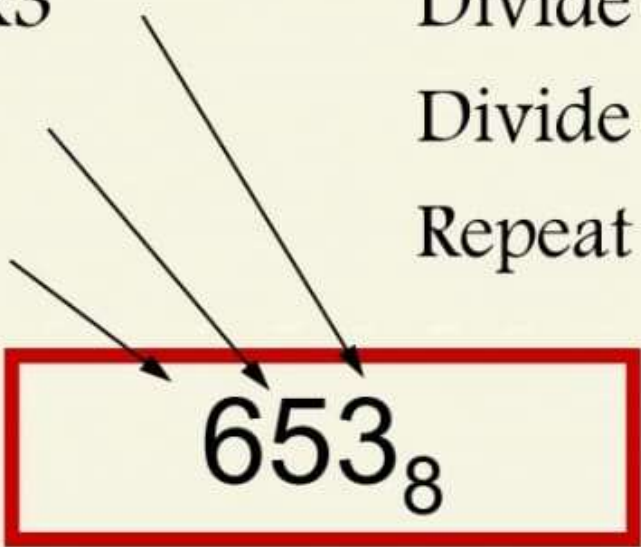
$$53 / 8 = 6 \text{ R}5$$

$$6 / 8 = 0 \text{ R}6$$

Divide by 8; R is LSD

Divide Q by 8; R is next digit

Repeat until $Q = 0$



653₈

Octal to Decimal Conversion

Convert 653_8 to its decimal equivalent.

Octal Digits



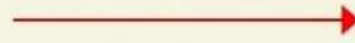
6 5 3
x x x

Positional Values



8^2 8^1 8^0

Products



$384 + 40 + 3$

427_{10}

Octal to Binary Conversion

Each octal number converts to 3 binary digits

Code
0 - 000
1 - 001
2 - 010
3 - 011
4 - 100
5 - 101
6 - 110
7 - 111

To convert 653_8 to binary, just substitute code:

6	5	3
↓	↓	↓
110	101	011

Hexadecimal Number System

- Base 16 system
- Uses digits 0-9 & letters A,B,C,D,E,F
- Groups of four bits represent each base 16 digit

Decimal	Hexadecimal
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	A
11	B
12	C
13	D
14	E
15	F

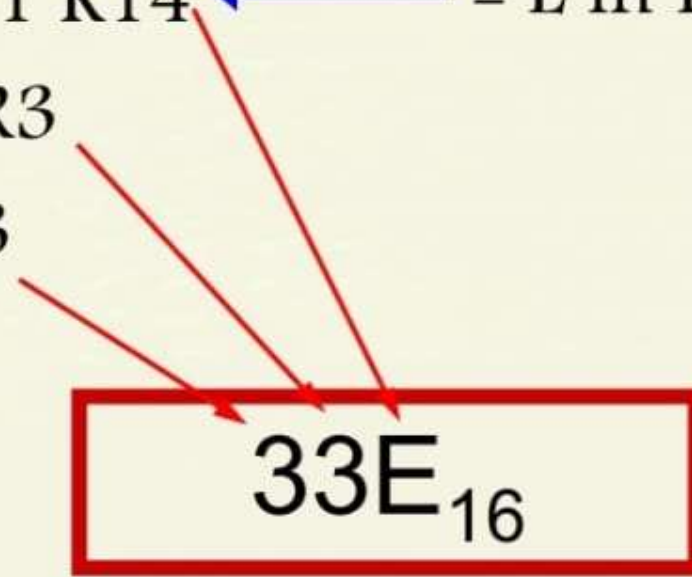
Decimal to Hexadecimal Conversion

Convert 830_{10} to its hexadecimal equivalent.

$$830 / 16 = 51 \text{ R}14 \quad \leftarrow = \text{E in Hex}$$

$$51 / 16 = 3 \text{ R}3$$

$$3 / 16 = 0 \text{ R}3$$



33E₁₆

Hexadecimal to Decimal Conversion

Convert $3B4F_{16}$ to its decimal equivalent.

Hex Digits	→	3	B	4	F
		x	x	x	x
Positional Values	→	16^3	16^2	16^1	16^0
Products	→	$12288 + 2816 + 64 + 15$			
		15,183₁₀			

Binary to Hexadecimal Conversion

- The easiest method for converting binary to hexadecimal is to use a **substitution code**
- Each hex number converts to 4 binary digits

Substitution Code

0000 = 0	0100 = 4	1000 = 8	1100 = C
0001 = 1	0101 = 5	1001 = 9	1101 = D
0010 = 2	0110 = 6	1010 = A	1110 = E
0011 = 3	0111 = 7	1011 = B	1111 = F

Substitution Code

Convert $010101101010111001101010_2$ to hex using the 4-bit substitution code :

0101	0110	1010	1110	0110	1010
5	6	A	E	6	A

56AE6A₁₆

Substitution Code

Substitution code can also be used to convert binary to octal by using 3-bit groupings:

010	101	101	010	111	001	101	010
2	5	5	2	7	1	5	2

25527152₈

